

GIT

Registration No. : 2025/CS/079

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```
[chamal@MacBook-Air: ~ $ git --version
git version 2.54.0
[chamal@MacBook-Air: ~/Desktop $ mkdir git-tutorial
[chamal@MacBook-Air: ~/Desktop $ cd git-tutorial
[chamal@MacBook-Air: ~/Desktop/git-tutorial $ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /Users/chamalinduware/Desktop/git-tutorial/.git/
[chamal@MacBook-Air: ~/Desktop/git-tutorial $
[chamal@MacBook-Air: ~/Desktop/git-tutorial $ echo "# My Practice Project" > README.md
[chamal@MacBook-Air: ~/Desktop/git-tutorial $ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    README.md

nothing added to commit but untracked files present (use "git add" to track)
[chamal@MacBook-Air: ~/Desktop/git-tutorial $ git add README.md
[chamal@MacBook-Air: ~/Desktop/git-tutorial $ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   README.md

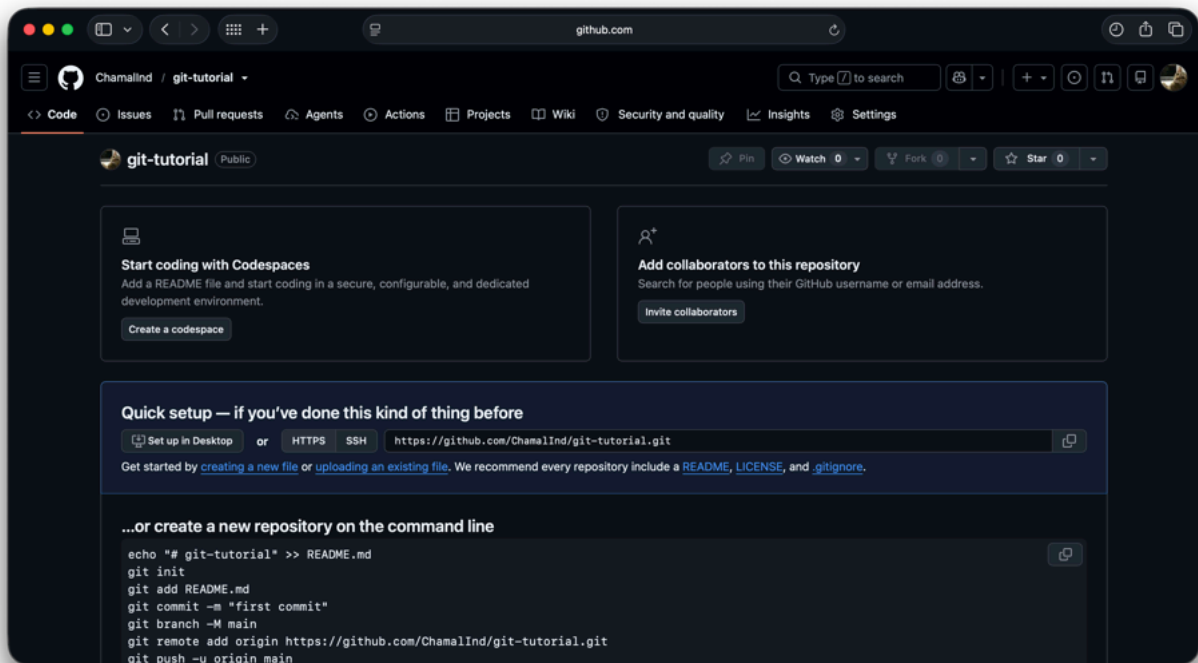
[chamal@MacBook-Air: ~/Desktop/git-tutorial $ git commit -m "Add README"
[master (root-commit) 15b72f7] Add README
 1 file changed, 1 insertion(+)
 create mode 100644 README.md
[chamal@MacBook-Air: ~/Desktop/git-tutorial $ git log
commit 15b72f73ea66210d6c59b4be39b2704e5ec3012b (HEAD -> master)
Author: Chamal Induwara <88148370+ChamalInd@users.noreply.github.com>
Date:   Fri Jul 10 15:26:51 2026 +0530

    Add README
```

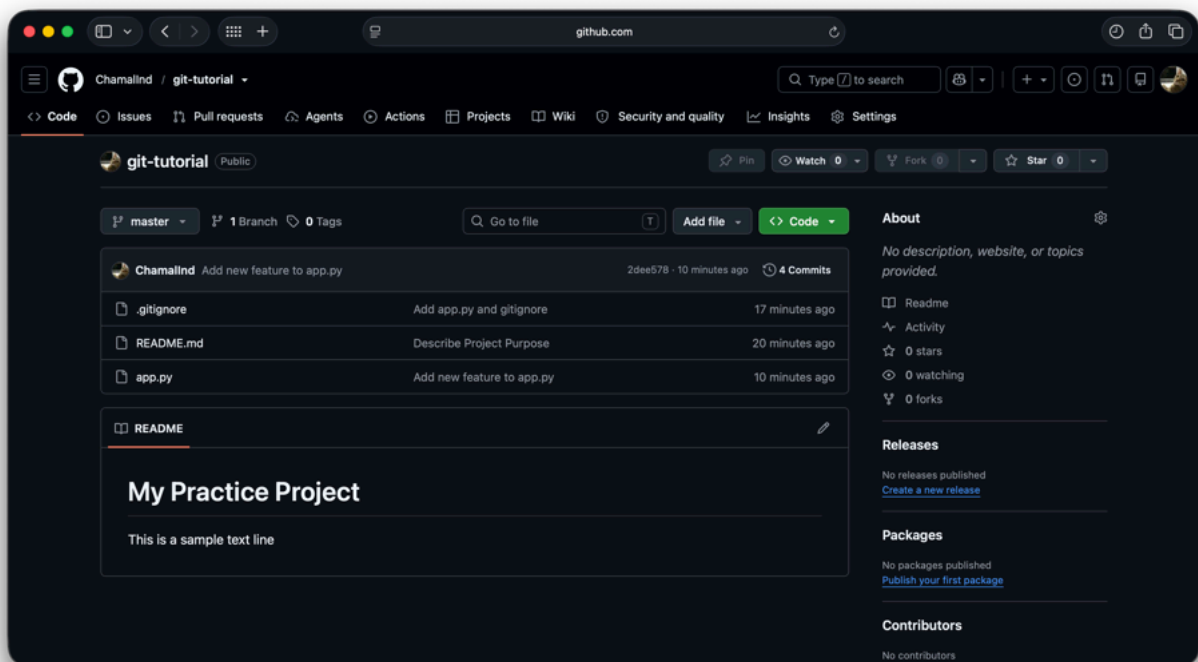
```

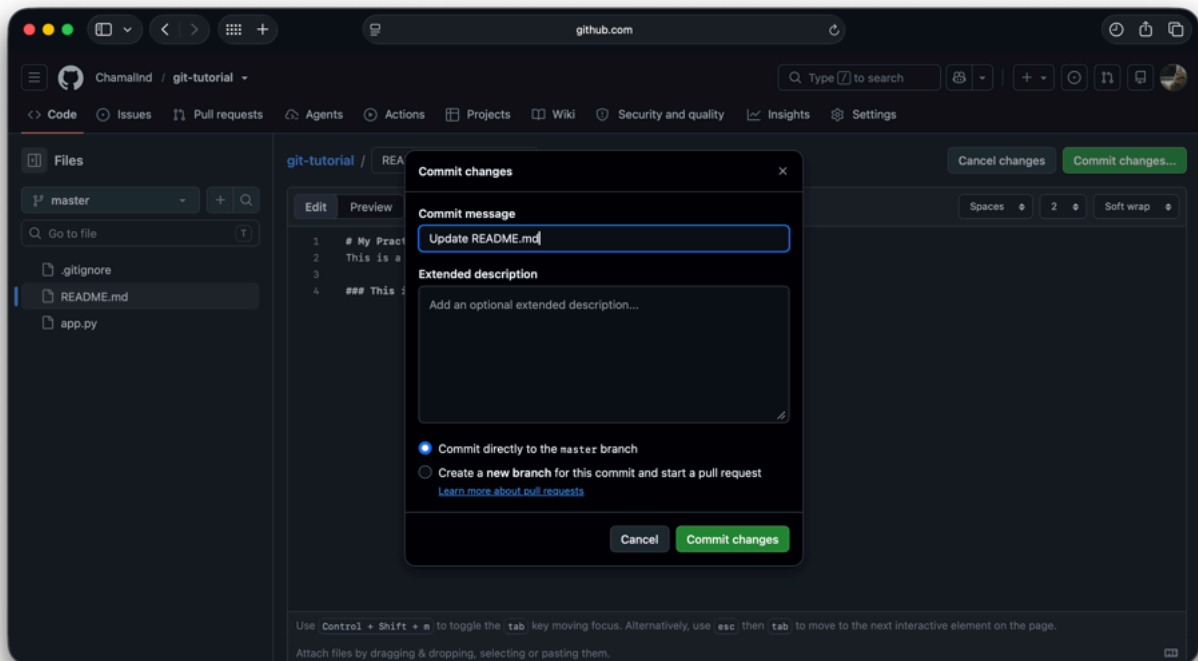
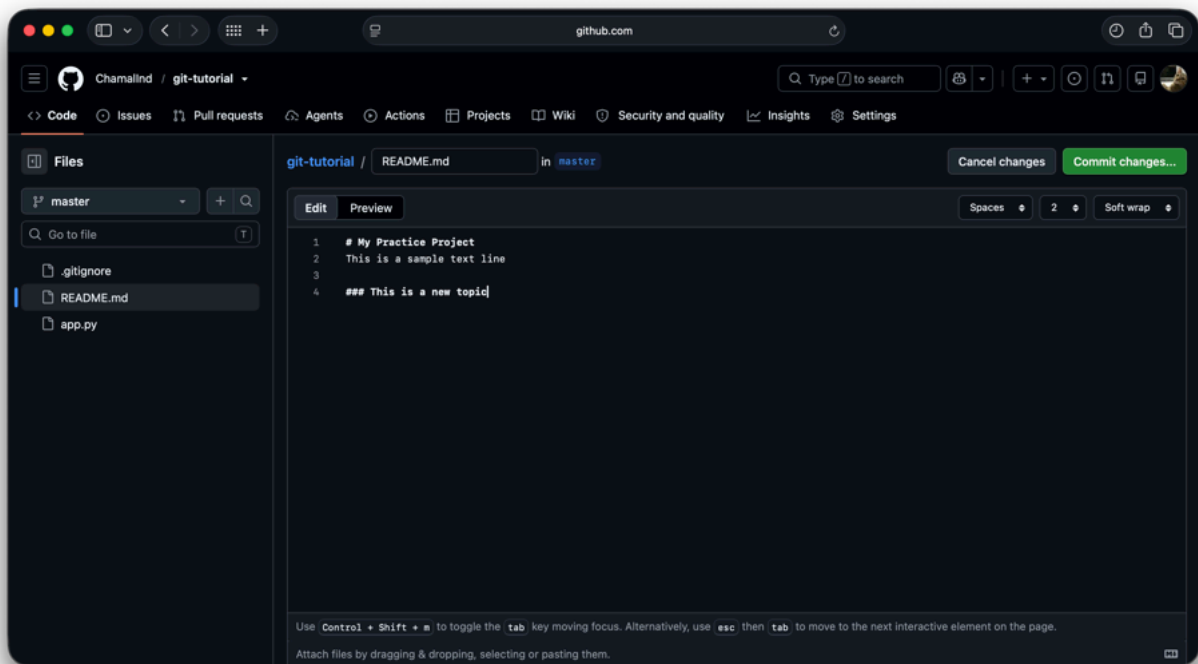
1 # My Practice Project
2 This is a sample text line
~
~
~
~
~
~
~
[chamal@MacBook-Air: ~/Desktop/git-tutorial $ vim README.md
[chamal@MacBook-Air: ~/Desktop/git-tutorial $ git diff
diff --git a/README.md b/README.md
index abac9c8..b0a3b69 100644
--- a/README.md
+++ b/README.md
@@ -1,2 @@
 # My Practice Project
+This is a sample text line
[chamal@MacBook-Air: ~/Desktop/git-tutorial $ git add README.md
[chamal@MacBook-Air: ~/Desktop/git-tutorial $ git commit -m "Describe Project Purpose"
[master eb8b524] Describe Project Purpose
1 file changed, 1 insertion(+)
[chamal@MacBook-Air: ~/Desktop/git-tutorial $ echo "print('hello')" > app.py
[chamal@MacBook-Air: ~/Desktop/git-tutorial $ echo "*.log" > .gitignore
[chamal@MacBook-Air: ~/Desktop/git-tutorial $ echo "__pycache__/" >> .gitignore
[chamal@MacBook-Air: ~/Desktop/git-tutorial $ git add .
[chamal@MacBook-Air: ~/Desktop/git-tutorial $ git commit -m "Add app.py and gitignore"
[master 2d84e9a] Add app.py and gitignore
2 files changed, 3 insertions(+)
create mode 100644 .gitignore
create mode 100644 app.py
[chamal@MacBook-Air: ~/Desktop/git-tutorial $ git log --oneline
2d84e9a (HEAD -> master) Add app.py and gitignore
eb8b524 Describe Project Purpose
15b72f7 Add README
[chamal@MacBook-Air: ~/Desktop/git-tutorial $ git branch
* master
[chamal@MacBook-Air: ~/Desktop/git-tutorial $ git checkout -b add-feature
Switched to a new branch 'add-feature'
[chamal@MacBook-Air: ~/Desktop/git-tutorial $ echo "print('new feature')" >> app.py
[chamal@MacBook-Air: ~/Desktop/git-tutorial $ git add app.py
[chamal@MacBook-Air: ~/Desktop/git-tutorial $ git commit -m "Add new feature to app.py"
[add-feature 2dee578] Add new feature to app.py
1 file changed, 1 insertion(+)
[chamal@MacBook-Air: ~/Desktop/git-tutorial $ git checkout master
Switched to branch 'master'
[chamal@MacBook-Air: ~/Desktop/git-tutorial $ cat app.py
print('hello')
[chamal@MacBook-Air: ~/Desktop/git-tutorial $ git merge add-feature
Updating 2d84e9a..2dee578
Fast-forward
 app.py | 1 +
1 file changed, 1 insertion(+)
[chamal@MacBook-Air: ~/Desktop/git-tutorial $ cat app.py
print('hello')
print('new feature')

```



```
chamal@MacBook-Air: ~/Desktop/git-tutorial $ git remote add origin https://github.com/ChamalInd/git-tutorial.git
chamal@MacBook-Air: ~/Desktop/git-tutorial $ git push -u origin master
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (13/13), 1.08 KiB | 1.08 MiB/s, done.
Total 13 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
To https://github.com/ChamalInd/git-tutorial.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
```





```

chamal@MacBook-Air: ~/Desktop/git-tutorial $ git pull
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 1023 bytes | 170.00 KiB/s, done.
From https://github.com/ChamalInd/git-tutorial
  2dee578..68ba0e1  master    -> origin/master
Updating 2dee578..68ba0e1
Fast-forward
 README.md | 2 ++
 1 file changed, 2 insertions(+)
chamal@MacBook-Air: ~/Desktop/git-tutorial $ echo "BADLINE" >> app.py
chamal@MacBook-Air: ~/Desktop/git-tutorial $ git add app.py
chamal@MacBook-Air: ~/Desktop/git-tutorial $ git commit -m "Bad Commit"
[master cf320f9] Bad Commit
 1 file changed, 1 insertion(+)
chamal@MacBook-Air: ~/Desktop/git-tutorial $ git revert HEAD
[master 662208c] Revert "Bad Commit"
 1 file changed, 1 deletion(-)

 1 Revert "Bad Commit"
 2
 3 This reverts commit cf320f9aa58b2411a205c7adba7b95faa556df25.
 4
 5 # Please enter the commit message for your changes. Lines starting
 6 # with '#' will be ignored, and an empty message aborts the commit.
 7 #
 8 # On branch master
 9 # Your branch is ahead of 'origin/master' by 1 commit.
10 #   (use "git push" to publish your local commits)
11 #
12 # Changes to be committed:
13 #   modified:   app.py
14 #
~
~
~
~
~
~
~
~
~
chamal@MacBook-Air: ~/Desktop/git-tutorial $ cat app.py
print('hello')
print('\nNew feature!')
chamal@MacBook-Air: ~/Desktop/git-tutorial $ echo "TEMP WORK" >> app.py
chamal@MacBook-Air: ~/Desktop/git-tutorial $ git stash
Saved working directory and index state WIP on master: 662208c Revert "Bad Commit"
chamal@MacBook-Air: ~/Desktop/git-tutorial $ git status
On branch master
Your branch is ahead of 'origin/master' by 2 commits.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
chamal@MacBook-Air: ~/Desktop/git-tutorial $ git stash pop
On branch master
Your branch is ahead of 'origin/master' by 2 commits.
  (use "git push" to publish your local commits)

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   app.py

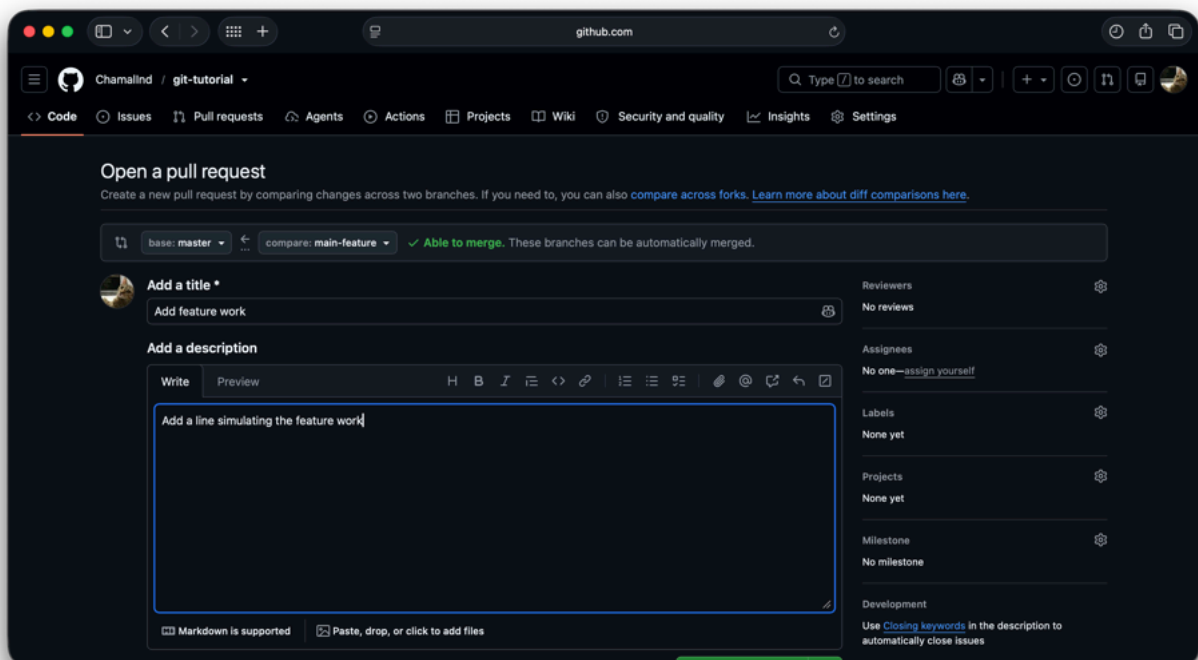
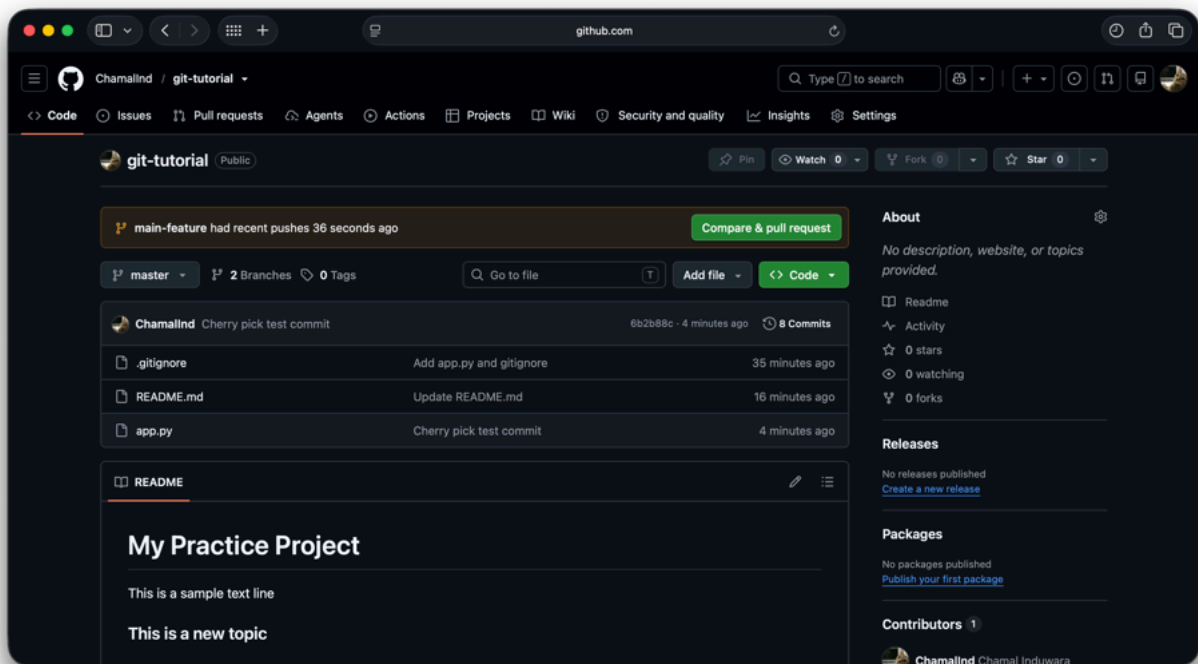
no changes added to commit (use "git add" and/or "git commit -a")
Dropped refs/stash@{0} (475188521f2cc661464789b9909402060045ec7f)

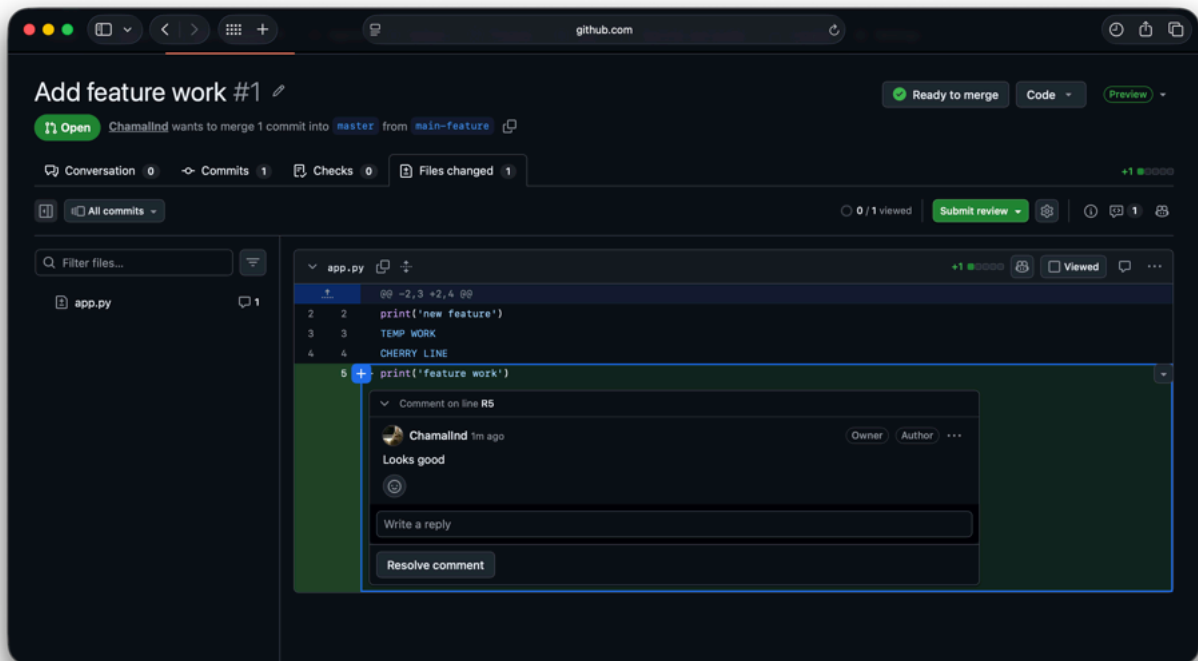
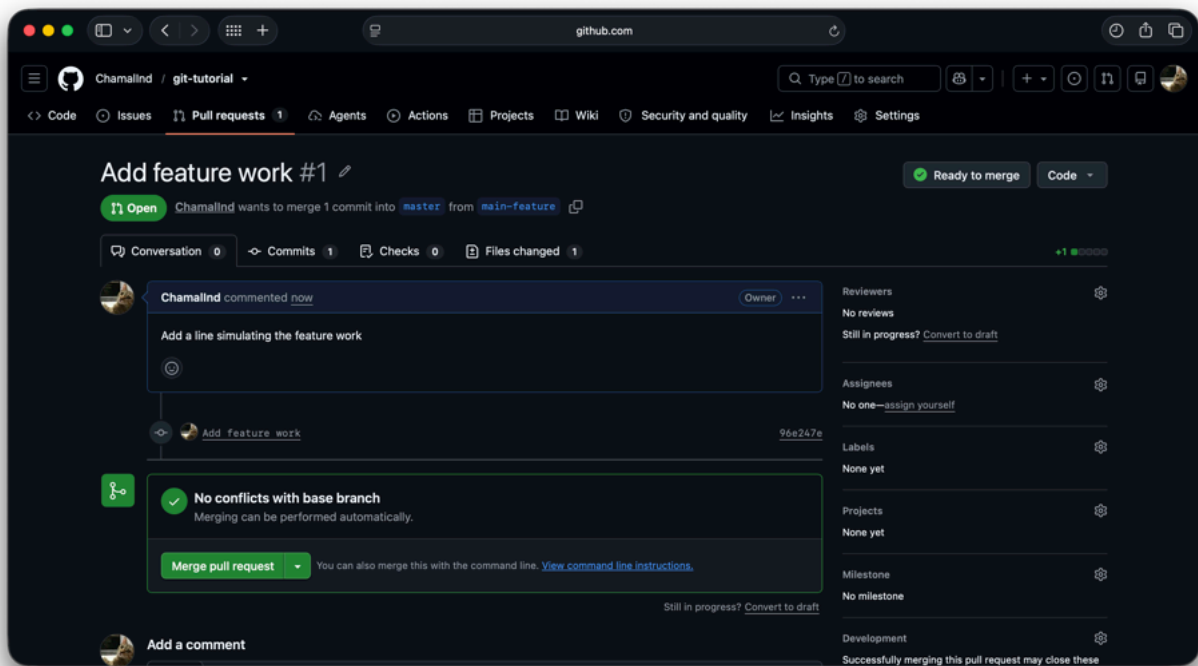
```

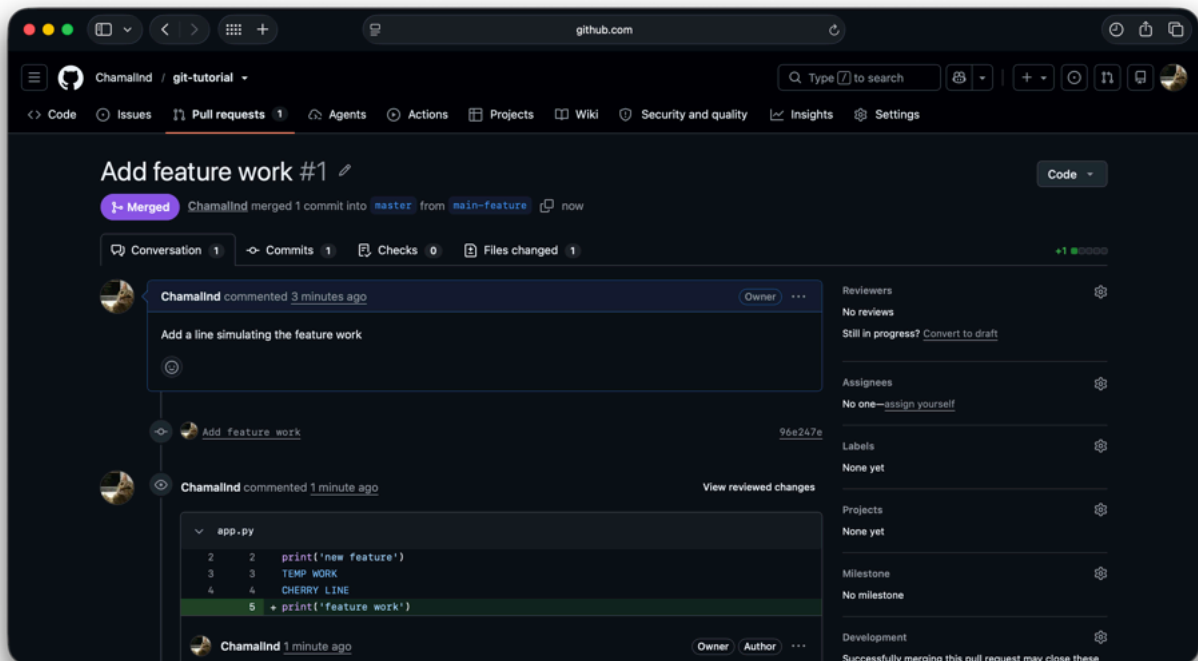
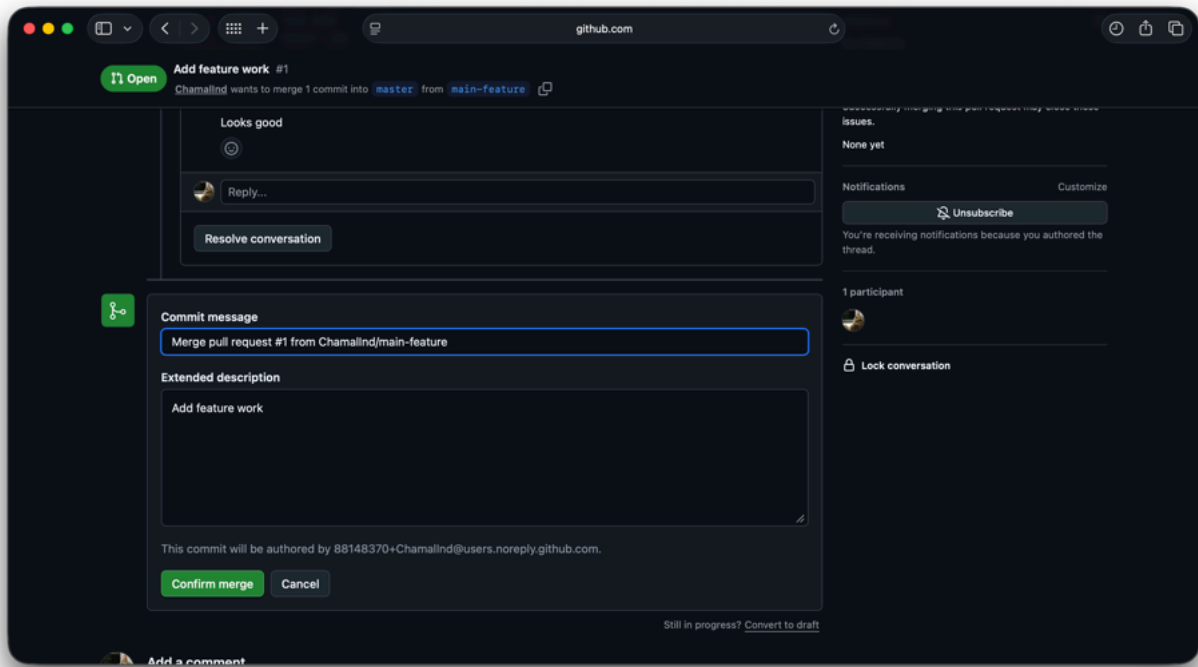
```

chamal@MacBook-Air: ~/Desktop/git-tutorial $ git checkout -b temp-branch
Switched to a new branch 'temp-branch'
chamal@MacBook-Air: ~/Desktop/git-tutorial $ echo "CHERRY LINE" >> app.py
chamal@MacBook-Air: ~/Desktop/git-tutorial $ git add app.py
chamal@MacBook-Air: ~/Desktop/git-tutorial $ git commit -m "Cherry pick test commit"
[temp-branch 986fc0c] Cherry pick test commit
1 file changed, 2 insertions(+)
chamal@MacBook-Air: ~/Desktop/git-tutorial $ git checkout master
Switched to branch 'master'
Your branch is ahead of 'origin/master' by 2 commits.
(use "git push" to publish your local commits)
chamal@MacBook-Air: ~/Desktop/git-tutorial $ cat app.py
print('hello')
print('new feature')
chamal@MacBook-Air: ~/Desktop/git-tutorial $ git log temp-branch --oneline
986fc0c (temp-branch) Cherry pick test commit
662208c (HEAD -> master) Revert "Bad Commit"
cf320f9 Bad Commit
68ba0e1 (origin/master, origin/HEAD) Update README.md
2dee578 Add new feature to app.py
2d84e9a Add app.py and gitignore
eb8b524 Describe Project Purpose
15b72f7 Add README
chamal@MacBook-Air: ~/Desktop/git-tutorial $ git cherry-pick 986fc0c
[master 6b2b88c] Cherry pick test commit
Date: Fri Jul 10 16:01:44 2026 +0530
1 file changed, 2 insertions(+)
chamal@MacBook-Air: ~/Desktop/git-tutorial $ cat app.py
print('hello')
print('new feature')
TEMP WORK
CHERRY LINE
chamal@MacBook-Air: ~/Desktop/git-tutorial $ git branch -d temp-branch
error: the branch 'temp-branch' is not fully merged
hint: If you are sure you want to delete it, run 'git branch -D temp-branch'
hint: Disable this message with "git config set advice.forceDeleteBranch false"
chamal@MacBook-Air: ~/Desktop/git-tutorial $ git branch -D temp-branch
Deleted branch temp-branch (was 986fc0c).
chamal@MacBook-Air: ~/Desktop/git-tutorial $ git push
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (7/7), 813 bytes | 813.00 KiB/s, done.
Total 7 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To https://github.com/ChamalInd/git-tutorial.git
68ba0e1..6b2b88c master -> master
chamal@MacBook-Air: ~/Desktop/git-tutorial $ git checkout -b main-feature
Switched to a new branch 'main-feature'
chamal@MacBook-Air: ~/Desktop/git-tutorial $ echo "print('feature work')" >> app.py
chamal@MacBook-Air: ~/Desktop/git-tutorial $ git add app.py
chamal@MacBook-Air: ~/Desktop/git-tutorial $ git commit -m "Add feature work"
[main-feature 96e247e] Add feature work
1 file changed, 1 insertion(+)
chamal@MacBook-Air: ~/Desktop/git-tutorial $ git push -u origin main-feature
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 339 bytes | 339.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'main-feature' on GitHub by visiting:
remote:   https://github.com/ChamalInd/git-tutorial/pull/new/main-feature
remote:
To https://github.com/ChamalInd/git-tutorial.git
* [new branch]    main-feature -> main-feature
branch 'main-feature' set up to track 'origin/main-feature'.

```







```

chamal@MacBook-Air: ~/Desktop/git-tutorial $ git checkout master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
chamal@MacBook-Air: ~/Desktop/git-tutorial $ git pull
remote: Enumerating objects: 1, done.
remote: Counting objects: 100% (1/1), done.
remote: Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (1/1), 904 bytes | 301.00 KiB/s, done.
From https://github.com/ChamalInd/git-tutorial
 6b2b88c..4482ac9 master    -> origin/master
Updating 6b2b88c..4482ac9
Fast-forward
 app.py | 1 +
 1 file changed, 1 insertion(+)
chamal@MacBook-Air: ~/Desktop/git-tutorial $ git branch -d main-feature
Deleted branch main-feature (was 96e247e).

```

